Coding Standard
Department of Computer Science
School of Engineering and Sciences
Tecnológico de Monterrey

The following defines the standards to be applied in every course, regardless of the programming language used.

Standards to be applied in all courses, concerning:

Identifier names

- Identifier names must be meaningful and convey information about their contents.
- Variable and function/method names must use *camelCase* notation (the first letter of the identifier must be lowercase; if the name is made up of several words, all of them except the first must begin with a capital letter (*descriptiveName*)).
- Constant names must use uppercase letters only; if the name is made up of several words, they must be separated by the "_" character (DESCRIPTIVE_NAME).
- Identifier names must not contain accents, diaereses or the letter ñ, even if the IDE accepts them.

Code structure

- The elements of a statement must be separated by a blank space.
- A numeric literal with decimals must have at least one digit before and one digit after the decimal point.
- Expressions must use only the parentheses that are required.
- A line of code must contain only one statement.
- Statements contained in a control structure or function/method must be indented one *Tab* with respect to the structure that contains them.
- The opening brace "{" must be placed on the same line as the header of the control structure or function/method.
- The closing brace "}" must be aligned with the left side of the control structure or function/method it delimits.
- Functions/methods or sections of code must be separated by a blank line.
- The code must include only variables that are used.

Comments

- Comments must be phrases that add value and help the programmer understand the code. Write meaningful comments and avoid comments that are obvious from reading the statement they refer to.
- The first lines of every source file must include at least the following comment block:
  - A brief description of the program.
  - A line with the author(s): ~~name(s)~~ and student ID(s). Student IDs are enough. Do not include names.
  - A line with the date of creation/modification.
- Functions/methods must always include comments before their definition.
- Function/method comments must contain a brief description of the purpose of:
  - the function,
  - each parameter, and
  - the return value.
- Comments may be included in sections of code whenever they are needed to state their objective.
- Unused lines of code must not be left behind as comments.

Class names

- Class names must use *PascalCase* notation (the first letter of each word must be uppercase (*DescriptiveName*)) and must be singular.

Methods/functions

- Each function/method must perform a single task and perform it completely.
- Do not use *break* or *continue* inside the statement block of a loop.
- Variables must be defined with an initial value.